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DATA NETWORKS LAB 1 (50)

Basic Network Information and Boolean Operations

Lab Objectives:

- Identify the basic network information required for computers to communicate and transmit data on the network.
- Execute network commands used to retrieve network-specific information.

Instructions:

- Use the workstations in the FCI labs or your laptop to complete the lab work.
- Ensure that your lab workstation is connected to the NUST Network.
- You may also use your laptop connected to the NUST Network.

Tools/Software Required:

- A computer running Windows OS
- Command-line interface (Command Prompt, PowerShell).

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Section 1: Basic Network Information

PART A [26]

1. Open the Command Prompt (CMD), execute the command ***ipconfig /all*** and record the following information of your workstation:

i. List all available network adapters on your computer and determine if they are logical or physical and if they are active or inactive. (1)

No	Network Adapter	Physical/Logical	Active/Inactive

Hint: There can only be one network interface card (NIC) that is both active and physical at a time. Other active network interfaces are usually logical (virtual) devices such as a loopback adapter or a VMware virtual adapter - or they may be disabled (e.g when you are connected via Wi-Fi while the wired Ethernet card is plugged in but not in use).

ii. Which of the above-listed network adaptors enabled your network Connection? (1)

_____

iii. Record the network information/ settings of this ethernet adapter: (4)

a) **Hostname:** _____

b) **IP Address:** _____

c) **MAC Address:** _____

d) **Subnet Mask:** _____

e) **Default Gateway:** _____

f) **DNS Server 1:** _____

g) **DNS Server 2:** _____

h) **DHCP Server:** _____

2. Explain the purposes of each setting when configuring network or TCP/IP. (8)

a) **Hostname:**

_____

_____

b) **IP Address:**

_____

_____

c) **MAC Address:**

_____

_____

d) **Subnet Mask:**

_____

_____

e) **Default Gateway:**

_____

_____

f) **DNS Server 1:**

_____

_____

g) **DNS Server 2:**

_____

_____

h) **DHCP Server:**

_____

_____

3. How is an IP address different from a MAC address, in terms of communication? (2)

_____

_____

_____

_____

4. How did your workstation obtain the IP Address, Subnet mask, and Default Mask (TCP/IP),

How can you tell? (2)

_____

_____

_____

5. Is it possible to change the IP assignment method you identified in Question 4 ? Explain how; action it and write down the steps. (3)

_____

_____

_____

6. Describe another method and write down the steps to retrieve the information in Question 1(ii) above without using the command line. (3)

_____

_____

_____

_____

_____

_____

9. Type **ipconfig** in the command prompt. Do the results look the same as the **ipconfig /all** command above? What is the difference and explain why. (2)

_____

_____

_____

PART B [14]

1. Using your computer's network information recorded in **Part A; 1(ii)**

a) Compare **ALL** your network information labelled **a - h** to the student next to you. Make sure the comparison is made block by block/octet by octet.

NB: ONLY compare results with students connected to the same network.

	Own Information	Other Student
a) **Hostname:**	_____	_____
b) **IP Address:**	_____	_____
c) **MAC Address:**	_____	_____
d) **Subnet Mask:**	_____	_____
e) **Default Gateway:**	_____	_____
f) **DNS Server 1:**	_____	_____
g) **DNS Server 2:**	_____	_____
h) **DHCP Server:**	_____	_____

b) Which of the above settings are the same as the other student? Explain why they are the same. (2)

_____

_____

_____

_____

_____

_____

c) Which of the above settings are different from the other student? Explain why each setting is different and what are the possible issues that can arise if these settings are the same. (2)

_____

_____

_____

_____

_____

2.

a) Test network connectivity by pinging the following addresses (2)

ping 8.8.8.8

i) Is there network connectivity? State YES or NO. Explain how you can tell from the results whether the host is reachable or not. Refer to specific messages shown in the command output

_____

_____

ping 192.192.168.10

ii) Is there network connectivity? State YES or NO. Explain how you can tell from the results whether the host is reachable or not. Refer to specific messages shown in the command output

_____

_____

b) On your command prompt, type **arp –a** to display the MAC address table and list the MAC address that your computer has learned. (1)

_____

_____

_____

_____

c) Why do computers need to learn each other's MAC addresses? (1)

_____

_____

_____

d) On the command prompt, **ping** the other student's computer (Part B(1a)) and issue the **arp –a** command again. The default gateway's MAC address should be in this list, as should the MAC address of the computer you just pinged; why? ***(Command: Ping [IP Address])*** (1)

_____

_____

_____

_____

e) What is the purpose of the **arp**, **arp -a** and **ping** commands? (3)

_____

_____

_____

_____

Execute the following commands, record the output and explain the function of each command: (4)

- **tracert www.google.com**

  _____
  _____
  _____

- **nslookup 8.8.8.8**

  _____
  _____
  _____

Section 3: Boolean Operations – Self Study [10]

The skill to convert between the number systems used in networking is extremely important. For additional practice, complete the following by converting the given value to the other number system.

NB: Boolean operations are covered in IT mathematics. If you did not do binary calculations before, study this topic, and complete the lab.

1. Convert the following IPv4 address and subnet mask from Decimal to Binary. Convert each octet separately.

IP Address:

192.168.50.100 → _____

Subnet Mask:
255.255.248.0 → _____

2. Convert the following Binary values to Decimal. Convert each octet separately and show your working.

IP Address (Binary):
10101100.00010000.01010000.00010100 → _____

Subnet Mask (Binary):
11111111.11111111.11111100.00000000 → _____

--End of Lab--